



Yong Chang Go

Software Engineer — Semiconductor Manufacturing

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About

Software engineer with 3+ years of experience building production automation systems in semiconductor manufacturing — from real-time SPC watchdogs and defect management platforms to MES-integrated lot tracking tools. Focused on reducing manual engineering effort through end-to-end system delivery.

Experience

Software Engineer, Data Analysis Platform

Jun 2025 — Present

VisionPower Semiconductor Manufacturing Company (VSMC)

- Build and maintain a **KLA Map-to-Map Data Watchdog** in Python — an automated SPC pipeline that monitors active lots, calculates DC values via formula-driven SQL against KLA inspection data, and submits results to MES with Oracle audit logging and automated retry on failure
Impact: Processing ~10 lots/hour autonomously, saving an estimated 1.5–2 engineering hours per hour of operation
- Maintain and enhance **VEDA**, a semiconductor engineering data analysis platform in C# (.NET WinForms + ASP.NET Web Services) with a 3-tier architecture, featuring cross-data-source correlation analysis (LQC, WAT, CP, SP, Lot History, FDC) with dynamic field mapping, and an interactive charting engine (trend, box plot, histogram, scatter, matrix) built on MS Chart Controls
Impact: Enabling engineers to identify yield-impacting equipment and process parameters across fab sites
- Maintain and enhance a full-stack **Defect Management System (DMS)** in C# (.NET/WinForms) — a config UI (SetupUI) with RBAC, a defect review UI with real-time Oracle WIP queries for lot hold/track-out/highlight operations, and an automated background service (Spec Check AP) triggering highlight actions via NATS messaging
Impact: Reducing manual review effort and improving defect response consistency across KLA, OM, SEMV, and ADC inspection equipment types

Software Developer, Manufacturing Execution Systems

Feb 2023 — Jun 2025

United Microelectronics Corporation (UMC), Singapore Branch

- Migrated and enhanced the **eRack Management System** from a legacy ASP.NET web application to C#/.NET Windows Forms with Oracle SQL and multi-threading for real-time MES sync via transaction calling
Impact: Significantly reduced FOUP check-in/check-out time and improved overall usability
- Built an **Automated WAT Rework and Backup System** in C# integrated with Aegis and ADO.NET, automating lot verification, hold code management, and parameter modifications
Impact: Drastically increased lot processing throughput compared to manual workflow, reducing issue-checking time through automated alarm email notifications
- Migrated the **Reticle Management System** from VB to C#/ASP.NET Web Forms, implementing AJAX partial updates and session management
Impact: Improved application performance and boosted daily operation efficiency
- Maintained and enhanced existing MES applications, collaborating with production staff to diagnose issues and deploy fixes with minimal downtime

Technical Skills

LANGUAGES

C# Python SQL
HTML/CSS/JavaScript
VB.NET SAS

FRAMEWORKS

.NET Framework
ASP.NET Web Forms
Windows Forms ADO.NET

TOOLS & PLATFORMS

Git Gitea Azure DevOps
DBeaver
Oracle SQL Developer NATS

DATABASES

Oracle

DOMAIN EXPERTISE

Semiconductor Manufacturing
MES Integration
Engineering Data Analysis
SPC Data Pipelines
Process Automation

Education

Bachelor of Information Technology (Honours) — Artificial Intelligence

Jun 2018 — Sep 2022

Multimedia University (Malacca Campus), Malacca

Second Class Upper (Honours)